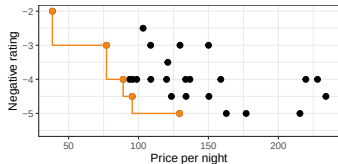
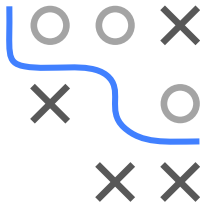


Optimization in Machine Learning

Multi-criteria Optimization Introduction



Learning goals

- What is multi-criteria optimization?
- Motivating examples
- Pareto optimality

INTRODUCTORY EXAMPLE

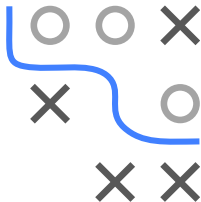
Often we want to solve optimization problems concerning several goals.

General applications:

- Medicine: maximum effect, but minimum side effect of a drug.
- Finances: maximum return, but minimum risk of an equity portfolio.
- Production planning: maximum revenue, but minimum costs.
- Booking a hotel: maximum rating, but minimum costs.

In machine learning:

- Sparse models: maximum predictive performance, but minimal number of features.
- Fast models: maximum predictive performance, but short prediction time.
- ...

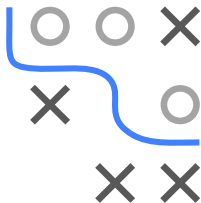


INTRODUCTORY EXAMPLE

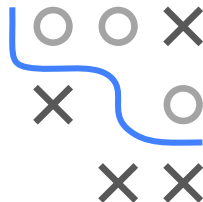
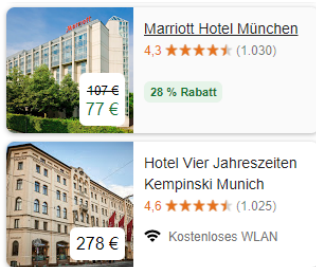
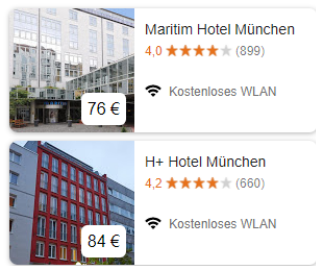
Example: Choose the best hotel by maximizing ratings subject to a maximum price per night. **Problems:**

- The result depends on how we select the maximum price; different price bounds give different solutions.
- We could also choose a minimum rating and optimize the price.
- The more objectives we include, the harder it gets to fix cutoffs like these.

Goal: Find a more general approach to solve multi-criteria problems.



INTRODUCTORY EXAMPLE



When booking a hotel: find the hotel with

- minimum price per night (**costs**), and
- maximum user rating (**performance**).

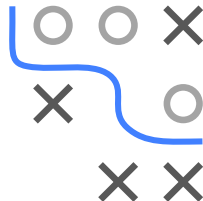
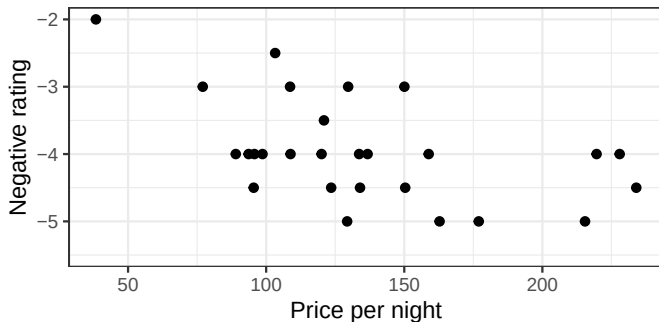
Since our standard is to minimize objectives, we minimize negative ratings.

INTRODUCTORY EXAMPLE

The objectives often conflict with each other:

- Lower price $\rightarrow$ usually lower hotel rating.
- Better rating $\rightarrow$ usually higher price.

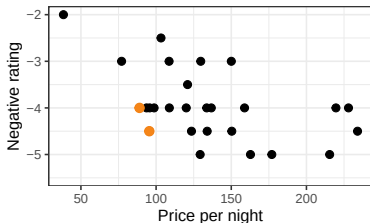
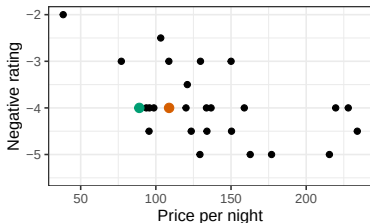
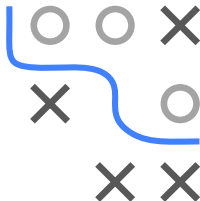
Example: (negative) average rating by hotel guests (1–5) vs. average price per night.



INTRODUCTORY EXAMPLE

We cannot directly establish a total order on the objective space:

- Left: A hotel with rating 4 for 89 Euro ($f^{(1)} = (89, -4.0)$) is better than a hotel for 108 Euro with the same rating ($f^{(2)} = (108, -4.0)$).
- Right: How to decide if $f^{(1)} = (89, -4.0)$ is better or worse than $f^{(2)} = (95, -4.5)$?
- How much is one *rating point* worth?



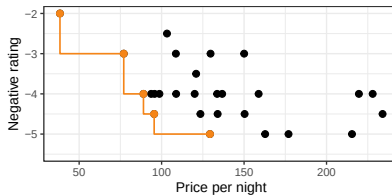
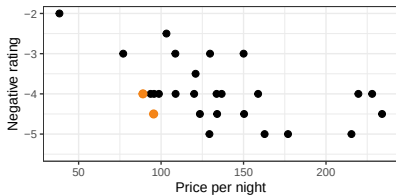
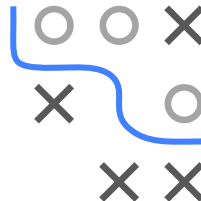
Let $f = (\text{price}, -\text{rating})$. The candidate $f^{(1)} = (89, -4.0)$ dominates $f^{(2)} = (108, -4.0)$. That is, $f^{(1)}$ is at least as good in every dimension, and strictly better in one dimension. Hence $f^{(1)} \prec f^{(2)}$. But for $f^{(1)} = (89, -4.0)$ and $f^{(2)} = (95, -4.5)$ we cannot say either is strictly better.

HOW TO DEFINE OPTIMALITY?

- We call these **incomparable** (or *equivalent* in the Pareto sense):

$$f^{(1)} \not\preceq f^{(2)} \quad \text{and} \quad f^{(2)} \not\preceq f^{(1)}.$$

- All such non-dominated points form the **Pareto front**.



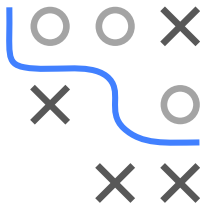
DEFINITION: MULTI-CRITERIA OPTIMIZATION

A **multi-criteria optimization problem** is defined by

$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x}) \Leftrightarrow \min_{\mathbf{x} \in \mathcal{S}} (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x})),$$

with $\mathcal{S} \subset \mathbb{R}^n$ and a multi-criteria objective function $f : \mathcal{S} \rightarrow \mathbb{R}^m$, $m \geq 2$.

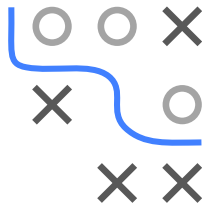
- **Goal:** minimize multiple target functions simultaneously.
- $(f_1(\mathbf{x}), \dots, f_m(\mathbf{x}))^\top$ maps each candidate $\mathbf{x}$ into the objective space $\mathbb{R}^m$.
- Typically no single best solution exists, as the objectives are often in conflict and cannot be totally ordered in $\mathbb{R}^m$.
- W.l.o.g. we always minimize.
- Also called multi-objective optimization, Pareto optimization, etc.



PARETO SETS AND PARETO OPTIMALITY

Definition: Given

$$\min_{\mathbf{x} \in \mathcal{S}} (f_1(\mathbf{x}), \dots, f_m(\mathbf{x})), \quad f_i : \mathcal{S} \rightarrow \mathbb{R}.$$



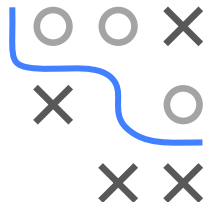
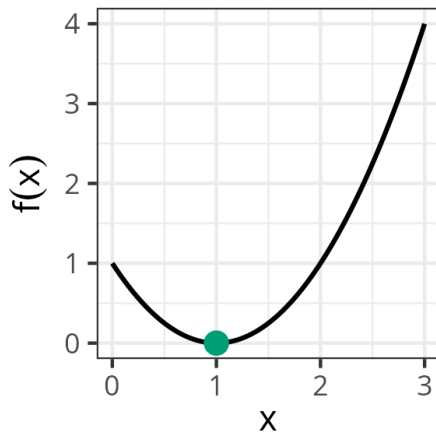
- A candidate $\mathbf{x}^{(1)}$ **(Pareto-)dominates** $\mathbf{x}^{(2)}$ if $f(\mathbf{x}^{(1)}) \prec f(\mathbf{x}^{(2)})$, i.e.:
 - 1 $f_i(\mathbf{x}^{(1)}) \leq f_i(\mathbf{x}^{(2)})$ for all i ,
 - 2 $f_j(\mathbf{x}^{(1)}) < f_j(\mathbf{x}^{(2)})$ for at least one j .
- A candidate $\mathbf{x}^*$ not dominated by any other candidate is **Pareto optimal**.
- The set of all Pareto optimal candidates is the **Pareto set**
 $\mathcal{P} = \{\mathbf{x} \in \mathcal{S} \mid \nexists \tilde{\mathbf{x}} : f(\tilde{\mathbf{x}}) \prec f(\mathbf{x})\}.$
- $\mathcal{F} = f(\mathcal{P})$, the **Pareto front**, is the image of that set in objective space.

EXAMPLE: ONE OBJECTIVE FUNCTION

We consider

$$\min_x f(x) = (x - 1)^2, \quad 0 \leq x \leq 3.$$

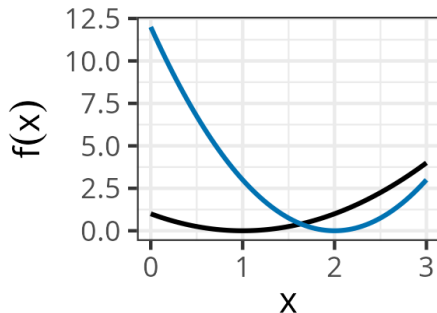
The optimum is at $x^* = 1$.



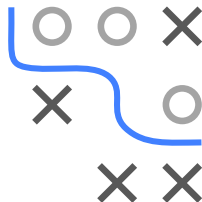
EXAMPLE: TWO TARGET FUNCTIONS

We extend the above problem to two objectives $f_1(x) = (x - 1)^2$ and $f_2(x) = 3(x - 2)^2$. Thus

$$\min_x f(x) = (f_1(x), f_2(x)), \quad 0 \leq x \leq 3.$$

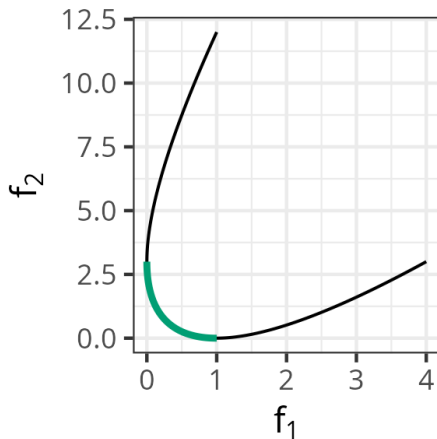


— f_1 — f_2

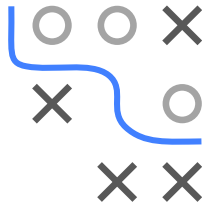


EXAMPLE: TWO TARGET FUNCTIONS

Visualizing both objectives: $(f_1(x), f_2(x))$ for all $x \in [0, 3]$ in the objective space:

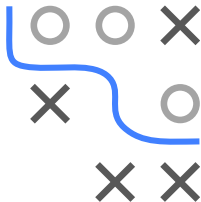


The Pareto front is shown in green.



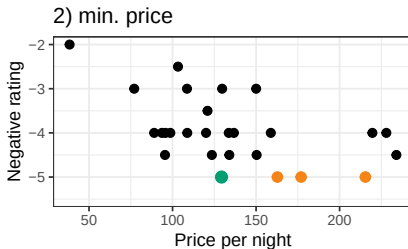
A-PRIORI VS. A-POSTERIORI

- The Pareto set is a set of equally valid solutions.
- Often, we only want a **single** solution in practice.
- Without extra info, there is no unique best solution: one must choose based on further criteria (cost constraints, business preference, etc.). Two overall strategies:
- **A-priori approach:** incorporate user preferences *before* the optimization.
- **A-posteriori approach:** first find (an approximation of) the entire Pareto set, then let the user choose.



A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path consists of the following cells: (0,0), (0,1), (1,1), (1,2), and (2,2). The cells (0,2), (1,0), and (2,0) are empty. The cells (0,1), (1,1), and (2,2) contain a grey 'X'. The cells (1,0) and (2,1) contain a grey circle.

Prior knowledge: The rating is more important than the price. Hence we optimize the rating first, then the price among those best ratings:



A-PRIORI PROCEDURE

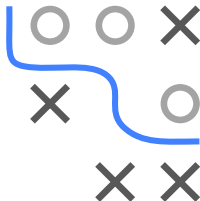
Lexicographic approach in general:

$$\begin{aligned}f_1^* &= \min_{\mathbf{x} \in \mathcal{S}} f_1(\mathbf{x}), \\f_2^* &= \min_{\substack{\mathbf{x} \in \mathcal{S}: \\ f_1(\mathbf{x}) = f_1^*}} f_2(\mathbf{x}), \\f_3^* &= \min_{\substack{\mathbf{x} \in \mathcal{S}: \\ f_1(\mathbf{x}) = f_1^*, f_2(\mathbf{x}) = f_2^*}} f_3(\mathbf{x}), \\&\vdots\end{aligned}$$

But different orderings yield different solutions.

Summary (a-priori):

- Implicit assumption: single-objective optimization is “easy.”
- We get only *one* solution, which depends strongly on the chosen weighting/order/etc.
- Varying these parameters systematically can produce several solutions, but may miss large parts of the non-dominated set.



A-POSTERIORI PROCEDURE

A-posteriori methods aim to

- find *all* optimal candidates (or a good approximation of them),
- let the user pick a single solution based on personal preferences or additional info (e.g. hotel location).

Hence, a-posteriori methods are more general.

Example: All Pareto-optimal hotels are shown on the left. User then picks from that subset on the right, based on hidden preferences (location, brand loyalty, etc.):

